Lab08_Mikhael

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```r
library(Replicate)
```

```python
import polars as pl

basics = pl.scan_csv(
    "title.basics0.tsv.gz",
    separator="\t",
    null_values="\\N",
    infer_schema_length=10000
)

ratings = pl.scan_csv(
    "title.ratings.tsv.gz",
    separator="\t",
    null_values="\\N"
).with_columns(
    pl.col("averageRating").cast(pl.Float64),
    pl.col("numVotes").cast(pl.Int64)
)

principals = pl.scan_csv(
    "title.principals0.tsv.gz",
    separator="\t",
    null_values="\\N"
)
```

1. Quais são os 5 filmes com as maiores notas (averageRating)? Apresente uma solução capaz de desempatar os filmes baseando-se no número de votos recebidos.

```python
filmes_avaliados = basics.join(
    ratings,
    on="tconst",
```

```
    how="inner"
)

top5 = (
    filmes_avaliados
    .filter(pl.col("titleType") == "movie")
    .sort(
        ["averageRating", "numVotes"],
        descending=[True, True]
    )
    .select(
        "primaryTitle",
        "startYear",
        "averageRating",
        "numVotes"
    )
    .head(5)
)

print(top5.collect(engine="streaming"))
```

```
shape: (5, 4)

 primaryTitle              startYear   averageRating   numVotes
 ---                       ---         ---             ---
 str                       i64         f64             i64

 Kaveri                    2024        10.0            1023
 Kurukku                   2024        10.0            451
 Jedal Dar Omghe 30 Metri  2009        10.0            142
 Sargashte                 2016        10.0            134
 Gorgeous Rascal           2024        10.0            115
```

2. Qual é o gênero mais frequente entre os filmes com nota maior que 8?

```
generos_frequentes = (
    basics
    .join(ratings, on="tconst", how="inner")
    .filter(
        (pl.col("titleType") == "movie") &
        (pl.col("averageRating") > 8)
```

```
    )
    .select("genres")
    .filter(pl.col("genres").is_not_null())
    .with_columns(
        pl.col("genres").str.split(",")
    )
    .explode("genres")
    .filter(
        (pl.col("genres") != "") &
        (pl.col("genres") != "\\N")
    )
    .group_by("genres")
    .agg(
        pl.len().alias("frequencia")
    )
    .sort(
        ["frequencia", "genres"],
        descending=[True, False]
    )
    .head(1)
)

print(generos_frequentes.collect(engine="streaming"))
```

```
shape: (1, 2)

 genres        frequencia
 ---           ---
 str           u32

 Documentary   10564
```

3. Quais são os 3 atores/atrizes que mais participaram de filmes com nota maior que 7.5?

```
top3atoresfilm = (
    principals
    .filter(
        pl.col("category").is_in(["actor", "actress"])
    )
    .join(
        basics.filter(pl.col("titleType") == "movie"),
```

```
        on="tconst",
        how="inner"
    )
    .join(
        ratings,
        on="tconst",
        how="inner"
    )
    .filter(pl.col("averageRating") > 7.5)
    .select("tconst", "nconst")
    .unique(subset=["tconst", "nconst"])
    .group_by("nconst")
    .agg(pl.len().alias("nfilmes"))
    .sort("nfilmes", descending=True)
    .head(3)
)

print(top3atoresfilm.collect(engine="streaming"))
```

```
shape: (3, 2)

 nconst      nfilmes
 ---         ---
 str         u32

 nm0004660   199
 nm0595934   155
 nm3183374   124
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-10-14 00:05:13 -03"
```